

A game on a directed container

Claude Opus 5 — for Robin Langer

Abstract

The repository **stream** contains a fighting game built on the *stream comonad*. Stream is the degenerate directed container: one shape, one future. This note defines the directing structure in general, gives pseudocode, and reports what happened when a game was designed on a non-degenerate one.

The short version: the game failed, twice, for reasons worth recording. The mathematics it was built on survives intact. And the place where the director *does* earn its keep turned out not to be the new game at all, but the old one — the renormalisation that the shipped game now runs on is a statement about the endomorphism monoid of a directed container.

1 Containers

A *container* is a pair $S \triangleleft P$ consisting of a set S of *shapes* and, for each shape $s : S$, a set $P s$ of *positions* in that shape. Shapes are templates; positions are the blanks in a template that data can be poured into. A container denotes a functor on **Set**:

$$\llbracket S \triangleleft P \rrbracket X = \sum_{s:S} X^{P s},$$

an element of which is a shape together with a labelling of its positions. A list is the container with $S = \mathbb{N}$ (the length) and $P s = \{0, \dots, s-1\}$. A stream is the container with $S = 1$ and $P * = \mathbb{N}$.

Containers denote functors, and functors alone. To get a *comonad* — to get **extract** and **duplicate**, and therefore **extend** — something more is needed, and that something is what a director supplies.

2 The director

Definition 1 (Ahman–Chapman–Uustalu). A *directed container* is a container $S \triangleleft P$ together with three operations

$$\begin{aligned} \downarrow : \prod s : S. P s &\rightarrow S && \text{the } \textit{subshape} \text{ at a position,} \\ o : \prod \{s : S\}. P s &&& \text{the } \textit{root} \text{ position,} \\ \oplus : \prod \{s : S\}. \prod p : P s. P (s \downarrow p) &\rightarrow P s && \textit{translation} \text{ into a global position,} \end{aligned}$$

satisfying two shape equations and three position equations:

$$s \downarrow o = s \tag{1}$$

$$s \downarrow (p \oplus p') = (s \downarrow p) \downarrow p' \tag{2}$$

$$p \oplus o = p \tag{3}$$

$$o \oplus p = p \tag{4}$$

$$(p \oplus p') \oplus p'' = p \oplus (p' \oplus p'') \tag{5}$$

I will call the triple $(\downarrow, o, \oplus)$ the *director* of the container, and $\downarrow$ itself the *directing map*: given where you are and a position you can see from there, it *directs* you to the world as seen from that position.

Remark. This is my naming, not the literature’s. Ahman, Chapman and Uustalu call $\downarrow$ the subshape operation, o the root and $\oplus$ translation, and do not use the word “director” anywhere in the paper. The polynomial-functor school does use *directions* for what containers call positions, which is the nearest standard usage. If “director” means something specific to you or to Kodamai that differs from Definition 1, this is the one place in the document that needs changing.

Read the laws as geometry. Law (1) says the subshape at the root is the whole shape — looking from where you already are changes nothing. Law (3) says the root of the subshape at p translates back to p itself, so the subshape given by a position really is *rooted* at that position. Law (2) says that going to p and then to p' lands on the same shape as going directly to $p \oplus p'$. Laws (3)–(5) are the monoid laws, except that the three positions live in three different sets, so it is a monoid only in a dependent sense. Laws (1)–(2) are the laws of a monoid *action*.

That last observation is the whole content of the definition. A directed container is a monoid-like structure of positions acting on shapes, where the monoid you get depends on where you are standing.

2.1 How the director is used: it is exactly a comonad

Theorem 2 (ACU). $\llbracket S \triangleleft P \rrbracket$ carries a comonad structure precisely when $S \triangleleft P$ carries a director.

The comonad is built from the director directly:

$$\begin{aligned}\varepsilon(s, f) &= f(o) \\ \delta(s, f) &= (s, \lambda p. (s \downarrow p, \lambda p'. f(p \oplus p')))\end{aligned}$$

ε reads the label at the root: *what is true here*. δ replaces the label at each position p by the entire structure re-rooted at p : *the world as it looks from there*. The type-checking of δ is exactly laws (1)–(2), and its comonad laws are exactly (3)–(5). And then

$$\text{extend} : (\llbracket S \triangleleft P \rrbracket A \rightarrow B) \rightarrow \llbracket S \triangleleft P \rrbracket A \rightarrow \llbracket S \triangleleft P \rrbracket B, \quad \text{extend } g = \llbracket S \triangleleft P \rrbracket g \circ \delta,$$

which runs a rule that needs to see a whole neighbourhood at *every* position at once. That is the operation `fmap` cannot express, and it is the reason the structure is worth having.

2.2 Stream is the degenerate case

Take $S = 1$, $P * = \mathbb{N}$, $s \downarrow p = s$, $o = 0$, $p \oplus p' = p + p'$. Laws (1)–(2) are trivial because the shape never changes; laws (3)–(5) are $p + 0 = p$, $0 + p = p$ and associativity of $+$. Then $\llbracket S \triangleleft P \rrbracket X = X^{\mathbb{N}}$ is streams, ε is `head`, and δ is `tails`. The file `StreamComonad.hs` prints δ applied to $0, 1, 2, \dots$ and observes that it is the addition table. That is law (2) with $\downarrow$ erased.

	stream	general directed container
shapes S	one	many: the world depends on where you stand
$s \downarrow p$	s , unchanged	a <i>different</i> world
$\oplus$	$+$, commutative	composition, not commutative
futures	exactly one	you choose

2.3 The director we use: a graph

Let G be a directed graph. Define

$$S = \text{vertices}, \quad Pv = \{\text{finite paths out of } v\}, \quad v \downarrow p = \text{target}(p), \quad o = \varepsilon, \quad p \oplus p' = p \cdot p'.$$

All five laws hold, and this is the free category on G . Now $\oplus$ is genuinely non-commutative and $\downarrow$ genuinely moves you.

This is not a coincidence of the example. Ahman and Uustalu proved that *directed containers are precisely small categories*: objects are shapes, morphisms out of s are positions in $P s$, $\downarrow$ is codomain, o is the identity and $\oplus$ is composition. A director is a category wearing the notation of a data structure. Section 7 is where that fact does some work.

3 The world: a Rauzy graph

Fix an irrational slope α and let w be the Sturmian word of slope α . Its *Rauzy graph of order m* has one vertex per factor of length m — a *room* — and an edge $u \rightarrow v$ whenever $u = a\sigma$ and $v = \sigma b$ with $a\sigma b$ a factor, which *emits* the bit b .

Walking the graph emits a bit sequence, and **the room you stand in is the last m bits you emitted**. The rooms are not extra state; they are your own recent history.

Proposition 3. *The Rauzy graph of order m of a Sturmian word has exactly $m + 1$ rooms and $m + 2$ doors, and exactly one room has two doors, whose bits differ.*

Checked in `Rauzy.hs`, 40 slopes $\times$ 6 orders, 240/240 on each count. The $m + 1$ is the Sturmian complexity function $p(m) = m + 1$: the “one split at every window length” of `splitting.pdf` is **one branching room**. Depending on whether the right- and left-special factors coincide, the graph is a *theta* (two arcs reconverging) or a *figure-eight* (two loops meeting at a point).

4 The game that was designed, and why it failed

The design: you are inside the schedule rather than watching it go past. Each beat you choose a door; the bit it emits is the monster’s action that beat. Combat is unchanged from `Game.hs`. Most rooms have one door — no choice, exactly the old game. One room has two.

It failed twice, and both failures are worth more than the design was.

4.1 The fiction would not go into words

Asked how a monster can have doors, the answer is that it cannot. “Rooms” and “doors” were imported from a *picture* of the graph without checking that the metaphor survived the transplant. The vertices are not places; they are *situations* — the last m beats of the exchange — and the edges are a positioning choice that constrains what the monster can do next.

4.2 Stated plainly, the mechanic died

Once the fiction is stated properly — “you choose whether the monster strikes you” — the next question is immediate: why would you ever choose to be struck? The intended answer was a trade: the free hit now versus the density of the loop it commits you to. Measured over 400 slopes at each of five orders:

does taking the beat that is open <i>now</i> also give the denser loop?	
$m = 3, 5, 8, 13, 21$	1993/1993

They never disagree. The greedy step is always also the better long-run loop, so the one decision in the world has a dominant answer requiring no lookahead at all. What remained was an information problem — find out which door is open — with no strategy in it.

Remark. An earlier draft of this document called the reference player “Karp” and described the branch as a choice between convergents. Both were wrong, and the dominance result above is why. Plain greed is optimal; the max-mean-cycle framing was decorative. That claim is withdrawn.

5 Pseudocode

5.1 The director, as an interface

```
type Shape
type Position          -- indexed by a shape: Position s

class DirectedContainer where
  down  :: Shape -> Position s -> Shape          -- s /> p      the directing map
  root  :: Position s                             -- o
  plus  :: Position s -> Position (down s p) -> Position s -- p <+> p'

-- LAWS
--   down s root      == s           plus p root      == p
--   down s (plus p p') == down (down s p) p'
--   plus root p       == p
--   plus (plus p p') p'' == plus p (plus p' p'')
```

5.2 The comonad it generates

```
data Den x = Den Shape (Position -> x)          -- [/ S </ P /] x

extract :: Den x -> x
extract (Den s f) = f root                       -- what is true HERE

duplicate :: Den x -> Den (Den x)
duplicate (Den s f) =
  Den s (\p -> Den (down s p)                    -- the world re-rooted at p ...
          (\p' -> f (plus p p'))))               -- ... relabelled by translation

extend :: (Den a -> b) -> Den a -> Den b
extend g = fmap g . duplicate                   -- run g at EVERY position at once
```

Specialising to $S = 1$, $\text{down } s p = s$, $\text{root} = 0$, $\text{plus} = (+)$ turns `duplicate` into `tails` and `plus p p'` into `p + p'` — the addition table.

5.3 Building the world, and the fight

```
sturmian alpha rho =
  [ floor ((k+1)*alpha + rho) - floor (k*alpha + rho) | k <- [0..] ]

rauzy m w = groupByRoom [ (window m i, (w !! (i+m), window m (i+1)))
                          | i <- [0 .. prefix] ]
  where window m i = take m (drop i w)

fight graph start playerHP monsterHP policy =
```

```

loop start playerHP monsterHP emptyMap
where
  loop room php mhp known
  | mhp <= 0 = WIN
  | php <= 0 = DEAD
  | otherwise =
    let doors      = doorsOf graph room
        (i, attack) = policy room known      -- the coKleisli arrow
        (bit, next) = doors !! i             -- <-- this IS room /> p
        php'       = if attack and bit == STRIKE then php - 5 else php
        mhp'       = if attack and bit == OPEN  then mhp - 2  else mhp
    in loop next php' mhp' (insert (room, i) bit known)

```

`doors !! i` is the directing map applied to a length-one position, and `known` is a *partial* element of $\llbracket S \triangleleft P \rrbracket \text{Bit}$: a partial function on positions, defined exactly on the doors already walked. Scouting enlarges its domain.

6 What was measured

All figures 300 fights per cell, α uniform in $[0.25, 0.75]$.

6.1 The mathematics that survives

Proposition 4. *The open-densities of the two cycles through the branching room are the best rational approximations to $1 - \alpha$ from above and from below, among fractions with denominator at most $m + 1$.*

Agreement: 300/300 at $m = 3, 5, 8, 13$ and 299/300 at $m = 21$. Worked cases at $m = 8$:

α	$1 - \alpha$	cycle densities	best approximations
0.3026	0.6974	5/7, 2/3	5/7 and 2/3
0.3490	0.6510	5/8, 2/3	2/3 and 5/8
0.6536	0.3464	1/3, 3/8	3/8 and 1/3
0.7000	0.3000	1/3, 2/7	1/3 and 2/7

Remark (a correction). I first stated this with denominator at most m , and it failed on roughly 30% of slopes. Every failure had denominator exactly $m + 1$ — unsurprising, since the graph has $m + 1$ rooms and no simple cycle can be longer. With the corrected bound the agreement is 1499/1500.

6.2 The design that does not

Even before the dominance result, the decision was worth almost nothing at any usable map size. Mean damage over 120 beats with no HP pool, so that nothing hides behind a win threshold:

m	first arc	scouts both	oracle	arc gap Δd	doors differ by
3	104.1	131.3	137.0	0.1389	33.3 dmg
5	112.2	123.0	130.7	0.0854	20.5 dmg
8	114.8	114.5	125.0	0.0398	9.6 dmg
13	109.0	107.4	122.7	0.0200	4.8 dmg
21	103.7	98.1	121.4	0.0095	2.3 dmg

The gap decays roughly as $m^{-1.4}$, because the two return times are typically of very unequal size. At $m = 8$ — the size I had planned to ship — the two doors differ by under 10 damage in

a fight of 120.

6.3 And pursuit would not have saved it

A chase (you move two steps, the monster one) needs a board with somewhere to hide. Sturmian Rauzy graphs have exactly one fork at every order, by Proposition 3; and Arnoux–Rauzy words, whose complexity is $2m + 1$, also have exactly one, merely of out-degree three:

	$m = 5$	$m = 8$	$m = 13$	$m = 21$
Sturmian, branch vertices	1	1	1	1
out-degrees	one vertex of degree 2, rest degree 1			
Tribonacci, branch vertices	1	1	1	1
out-degrees	one vertex of degree 3, rest degree 1			

One fork per lap is one binary guess per lap, which is the decision structure the original game already has. Tribonacci’s three-way fork is the version that could not be straddled, and is where I would look if pursuit were wanted.

7 Where the director earns its keep

None of the above uses the director for anything a bare graph could not do: only length-one positions, a $\downarrow$ with no shape content, and **extend** never called once. But the identification *directed container* = *small category* has a consequence that is not decorative.

In a small category, the morphisms from an object to itself form a **monoid**. Translated back: the positions $p \in P s$ with $s \downarrow p = s$ are closed under $\oplus$, contain o , and are associative — by laws (1)–(5) and nothing else. For the Rauzy graph, the loops at the branching room are freely generated by the two simple cycles, because every loop there decomposes uniquely into them.

So a trajectory through the world, read *in the endomorphism monoid of the branching object*, is an infinite word on two letters: which loop was taken, each time round.

Proposition 5. *That word is Sturmian again.*

Checked at $m = 2, 3, 5, 8, 13$, six slopes each: complexity $k + 1$ in all 30/30 cases.

This is the classical renormalisation of a Sturmian word — passing to the derived word shifts the slope by the Gauss map, i.e. consumes one partial quotient of α — but stated in the container’s own language it says something sharper: **the directed container contains a copy of itself, sitting in the endomorphism monoid of its one branching object.**

7.1 And this is what the shipped game now runs on

At $m = 2$ the branching room is the factor $.H$, and the loop taken is just the next beat. Robin’s reading of the game is then a set of theorems rather than heuristics: for $1/2 < \alpha < 2/3$ the word contains no $..$ and no HHH (0/300 each), so of the three length-2 contexts exactly one is undetermined:

$$H. \rightarrow H \text{ certain,} \quad HH \rightarrow . \text{ certain,} \quad .H \rightarrow ? \text{ the split.}$$

Together with his replacement of the fight clock by *chip damage* — a blocked strike costs 1, so stalling is paid for in the same currency as everything else — this gives a game with no dominant strategy and a ceiling a table cannot reach. 300 fights, $\alpha \in [0.52, 0.62]$, 60 HP each:

strategy	wins
attack every beat	0%
swing only after HH — never wrong, too slow to outrun the chip	0%
swing after any strike — takes the split blind	23%
memory-8 table on the raw beats (256 contexts)	82%
renormalising, depth 4 (16 contexts on the derived word)	92%
perfect prediction	100%

A 16-context recursive player beats a 4096-context memoriser, because it learns α 's next partial quotient instead of memorising more of the word. The memory-8 table flatlines at 82–83% no matter how deep it goes; the renormaliser does not, and Section 7 is why.

8 What a richer game would have to use

The test I should have applied at the start: **if the mechanic still works after you delete the director, the director was decoration**. Three parts of Definition 1 went unused, and each is a mechanic.

1. **$P s$ is all paths, not all steps**. Make a move a declared path $p \in P s$: commit k beats, blind for k , land at $s \downarrow p$. Law (5) is exactly what makes a combo mean one thing however you bracket it, and non-commutativity of $\oplus$ is the mathematical content of *initiative* — invisible in the stream case, where positions commute.
2. **$\downarrow$ can consume the world**. The worst thing in the current design is `beats = 120`, a constant with no mathematical content. In the non-empty-list director, $s \downarrow p = s - p$: moving does not relocate you, it *spends* the world. Then “out of time” is “ $s \downarrow p$ has no positions left”, and irreversibility is felt rather than asserted.
3. **extend is the verb no functor has**. Make it a consumable: spend a *read*, and a fixed rule runs over your whole forward cone at once. A game whose central action is **extend** does not typecheck without δ , and δ does not exist without laws (1)–(2). That is a mechanic the director cannot be deleted from.

And morphisms give the difficulty dial for free: “drop the last letter” is a functor $G_{m+1} \rightarrow G_m$, so bounded memory is a *quotient* directed container and levelling up is climbing one rung of a tower whose inverse limit is the Sturmian system itself.

9 Status

In the repository and verified: `Rauzy.hs` (Proposition 3, 240/240), `bench/Warren.hs` (Proposition 4, 1499/1500), `bench/Chip.hs` and `Game.hs` (the chip scoring and the renormalising player). Designed but not written: `Directed.hs`, checking the five laws exhaustively on a container where $\oplus$ is non-commutative.

Not built, deliberately: the Warren. Its mathematics is above; its mechanic has a dominant answer. The next thing actually worth measuring is a renormaliser of depth $d > 1$, recursing all the way down the continued fraction of α rather than stopping after one level.

Sources. D. Ahman, J. Chapman, T. Uustalu, *When is a container a comonad?*, LMCS 10(3:14), 2014, <https://lmcs.episciences.org/894/pdf> — Definition 1 is their Definition 3.1. D. Ahman, T. Uustalu, *Directed containers as categories*, MSFP 2016, <https://arxiv.org/pdf/1604.01187>.